

Network Intrusion Detection Experiment Results

Dataset Information

Dataset Size: 23145 rows
Duplicate Rows: 0
Features (Full): 15
Features (No Ports, IPs kept): 13

Model Performance (Full Features)

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9991	0.9991	0.9991	0.9991
Decision Tree	0.9991	0.9991	0.9991	0.9991
Random Forest	0.9991	0.9991	0.9991	0.9991
SVM	0.9981	0.9981	0.9981	0.9981
KNN	0.9991	0.9991	0.9991	0.9991
XGBoost	0.9991	0.9991	0.9991	0.9991
LightGBM	0.9991	0.9991	0.9991	0.9991

Model Performance (Without Ports, IPs kept)

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9972	0.9972	0.9972	0.9972
Decision Tree	0.9944	0.9944	0.9944	0.9944
Random Forest	0.9944	0.9944	0.9944	0.9944
SVM	0.9962	0.9962	0.9962	0.9962
KNN	0.9962	0.9962	0.9962	0.9962
XGBoost	0.9962	0.9963	0.9962	0.9962
LightGBM	0.9972	0.9972	0.9972	0.9972

Performance Comparison

	Recall (Full)	F1 Score (Full)	Accuracy (No Ports)	Precision (No Ports)	Recall (No Ports)	F1 Score (No Ports)	Accuracy (Full)
Logistic Regression	0.9991	0.9991	0.9972	0.9972	0.9972	0.9972	0.9991
Decision Tree	0.9991	0.9991	0.9944	0.9944	0.9944	0.9944	0.9991
Random Forest	0.9991	0.9991	0.9944	0.9944	0.9944	0.9944	0.9991
SVM	0.9981	0.9981	0.9962	0.9962	0.9962	0.9962	0.9981
KNN	0.9991	0.9991	0.9962	0.9962	0.9962	0.9962	0.9991
XGBoost	0.9991	0.9991	0.9962	0.9963	0.9962	0.9962	0.9991
LightGBM	0.9991	0.9991	0.9972	0.9972	0.9972	0.9972	0.9991

5-Fold Cross Validation

Model	CV Mean F1	CV Std
Logistic Regression	0.9912	0.0094
Decision Tree	0.9911	0.0091
Random Forest	0.9911	0.0091
SVM	0.9886	0.0086
KNN	0.9912	0.0095
XGBoost	0.9911	0.0091
LightGBM	0.992	0.0095